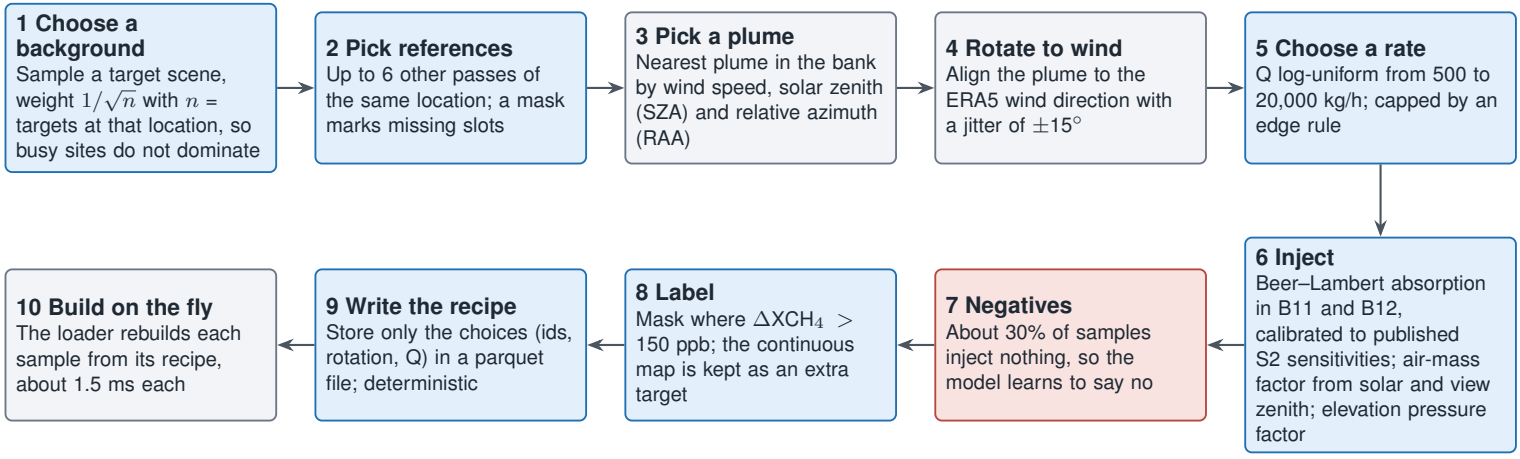
A synthetic sample is a recipe, not a file

Level 05 - synthetic recipe flow. From the pipeline notes of 7 and 13 October 2026.



What was produced (Windows run, 13 Oct 2026)
1,000,000 training recipes (70% positive, median rate about 3,149 kg/h) and 20,000 validation recipes (69.1% positive); validation materialised, training kept as recipes (about 20 GB, versus about 310 GB if written out). Background set: 112,688 images in 56,343 stacks. Plume bank: 279 plumes, 238,545 rows.

Licence flag
MethaneSET data and the MARS-S2L weights are CC BY-NC-SA 4.0: fine for the unit project, a blocker for any commercial use. Recorded in LICENSES.md.

Why recipes: 1 million rebuilt samples cost about 20 GB; storing them would cost about 310 GB. Determinism is the guard: the same recipe must give the same sample on every machine, and a test must prove it.